The 1st International Conference on Electrical, Electronics and Intelligent Systems — ICEEIS 2026

Footprint-Safe Adaptive Pivot–G^2 Path Smoothing for Differential-Drive AGVs in ROS 2/Nav2

Intelligent Control, Robotics, and Automation — Autonomous Systems (AMR, AGV, UAV)

Abstract—Global planners in ROS 2/Nav2 return collision-free polylines with abrupt heading changes. For differential-drive AGVs, preserving these vertices can induce frequent in-place rotations and high curvature variation, whereas unconstrained smoothing may cut corners and reduce obstacle clearance. This study addresses footprint-safe path post-processing in narrow warehouse passages. We propose Adaptive Pivot–G^2, which represents each corner either as an explicit in-place pivot or as a quintic Bézier transition that is geometrically G^2-continuous with adjacent line segments. An adaptive trim/radius search generates local candidates; curvature, wheel-motion, and swept-footprint checks reject infeasible candidates; and dynamic programming selects a globally compatible sequence without overlap between adjacent transitions. Evaluation used a paired benchmark of 7,200 path records comprising 60 scenarios distributed across seven maps, five global planners, eight post-processing methods, and three repetitions, with identical raw-path hashes within every comparison group. Adaptive Pivot–G^2 succeeded in 882/900 cases (98.0%). On the 882 jointly successful pairs, it reduced mean translational curvature energy by 92.1%, shortened path length by 0.8%, and increased minimum footprint clearance by 7.3% relative to the raw paths, with a mean processing time of 6.09 ms and zero footprint-collision samples. Eight representative closed-loop Gazebo runs across the seven environments all completed without collision-monitor intervention; the mean ground-truth and controller-estimated tracking RMSEs were 2.42 cm and 0.69 cm, respectively. The results indicate that Adaptive Pivot–G^2 provides real-time, footprint-aware path smoothing while preserving pivots where continuous cornering is infeasible. Hardware trials and repeated statistical testing remain future work.

Keywords—Differential-drive robot; automated guided vehicle; path smoothing; G^2 continuity; ROS 2/Nav2.